jahnavi1207@jahnavi1207-VirtualBox: ~/PES1UG24CS196-pes...

```
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b
9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe
9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ find .pes/objects -t
ype f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ git add object.c
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ git commit -m "Phase
 1: Implement object_write with header, SHA256, atomic write"
[main fbbd862] Phase 1: Implement object_write with header, SHA256, atomic write
 1 file changed, 96 insertions(+), 6 deletions(-)
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ git commit -m "Phase
 1: Add deduplication check in object_write"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
```

jahnavi1207@jahnavi1207-VirtualBox: ~/PES1UG24CS196-pes...

```
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b
9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe
9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ find .pes/objects -t
ype f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ git add object.c
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ git commit -m "Phase
 1: Implement object_write with header, SHA256, atomic write"
[main fbbd862] Phase 1: Implement object_write with header, SHA256, atomic write
 1 file changed, 96 insertions(+), 6 deletions(-)
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ git commit -m "Phase
 1: Add deduplication check in object_write"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
```

jahnavi1207@jahnavi1207-VirtualBox: ~/PES1UG24CS196-pes...

```
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ gcc -o test_tree tes
t_tree.o object.o tree.o index.o -lcrypto
.
bash: .: filename argument required
.: usage: . filename [arguments]
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes init
bash: ./pes: No such file or directory
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ make pes
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ find .pes/objective
-type f
find: '.pes/objective': No such file or directory
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ find .pes/objects -t
ype f
```

jahnavi1207@jahnavi1207-VirtualBox: ~/PES1UG24CS196-pes...

```
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ xxd .pes/objects/2a/
594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf | head -20
00000000: 626c 6f62 2031 3500 5465 7374 2069 6e74  blob 15.Test int
00000010: 6567 7269 5879 0a                        egriXy.
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ git add tree.c
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ git commit -m "Phase
 2: Add build_tree_level helper for recursive subtree construction"
[main 29dca22] Phase 2: Add build_tree_level helper for recursive subtree constr
uction
 1 file changed, 75 insertions(+), 5 deletions(-)
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ git commit -m "Phase
 2: Handle direct file entries as blob TreeEntry"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ git commit -m "Phase
 2: Recursively build subtrees for nested directory paths"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
```

jahnavi1207@jahnavi1207-VirtualBox: ~/PES1UG24CS196-pes...

```
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ rm -rf .pes
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ echo "hello world" >
 file1.txt
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ echo "goodbye world"
 > file2.txt
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes add file1.txt
file2.txt
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes status
Staged changes:
  (nothing to show)

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  README.md
  untracked:  test_objects
  untracked:  test_tree
  untracked:  test_sequence.sh
  untracked:  object.c
  untracked:  .gitignore
```

```
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ cat .pes/index
100644 e67ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc 17767697
93 8 bye.txt
100644 05c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43 17767697
66 12 hello.txt
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$
```

jahnavi1207@jahnavi1207-VirtualBox: ~/PES1UG24CS196-pes-vcs

```
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ rm -rf .pes
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ echo "Hello" > hello.txt
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes add hello.txt
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes commit -m "Initial commit"
[main 0dafab7] Initial commit
Committed: 0dafab754e73... Initial commit
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ echo "World" >> hello.txt
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes add hello.txt
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes commit -m "Add world"
[main d5e64b8] Add world
Committed: d5e64b809d57... Add world
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ echo "Goodbye" > bye.txt
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes add bye.txt
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes commit -m "Add farewell"
[main e0aab13] Add farewell
Committed: e0aab13a114f... Add farewell
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ./pes log
commit e0aab13a114f30031a4a854cc8b183e3e5dda2b8a3861d87067330ff7c2d5cd0
Author: Jahnavi R <PES1UG24CS196>
Date:   1776769815

    Add farewell

commit d5e64b809d57867e60736c503e7afb63c49b5e922080039e42ee321b58496a89
Author: Jahnavi R <PES1UG24CS196>
Date:   1776769785

    Add world

commit 0dafab754e73c3fef055c58ee78d7150aa1b1e315c2b6e69eb739bd27078fb4a
Author: Jahnavi R <PES1UG24CS196>
Date:   1776769757

    Initial commit

jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ find .pes -type f | sort
.pes/HEAD
```

jahnavi1207@jahnavi1207-VirtualBox: ~/PES1UG24CS196-pes...

```
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ find .pes -type f |
sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/0d/afab754e73c3fef055c58ee78d7150aa1b1e315c2b6e69eb739bd27078fb4a
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d5/e64b809d57867e60736c503e7afb63c49b5e922080039e42ee321b58496a89
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e0/aab13a114f30031a4a854cc8b183e3e5dda2b8a3861d87067330ff7c2d5cd0
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ cat .pes/refs/heads/
main
e0aab13a114f30031a4a854cc8b183e3e5dda2b8a3861d87067330ff7c2d5cd0
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ make test-integratio
n
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)
```

jahnavi1207@jahnavi1207-VirtualBox: ~/PES1UG24CS196-pes...

```
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ cat .pes/refs/heads/
main
e0aab13a114f30031a4a854cc8b183e3e5dda2b8a3861d87067330ff7c2d5cd0
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ make test-integratio
n
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
[main c156862] Initial commit
Committed: c156862ca0ab... Initial commit

Log after first commit:
commit c156862ca0ab39beaa573815d39b4ad49e9d0eb269fbd894d0d2df05b28ea8c9
Author: Jahnavi R <PES1UG24CS196>
```

jahnavi1207@jahnavi1207-VirtualBox: ~/PES1UG24CS196-pes-vcs

```
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
[main bdf433b] Initial commit
Committed: bdf433b86312... Initial commit

Log after first commit:
commit bdf433b863121575175810409a453904fcf5b170a88826c09874041868d2fd8b
Author: Jahnavi R <PES1UG24CS196>
Date:   1776770154

    Initial commit


--- Second Commit ---
[main 3842f48] Update file.txt
```

```
[main 3842f48] Update file.txt
Committed: 3842f48f44b8... Update file.txt

--- Third Commit ---
[main 536d031] Add farewell
Committed: 536d0316dfe4... Add farewell

--- Full History ---
commit 536d0316dfe49cc3931fd170f79e2a3f7cc06a26219a7c0fc4f4a02c98b75f55
Author: Jahnavi R <PES1UG24CS196>
Date:   1776770154

    Add farewell

commit 3842f48f44b8c0916a450c3f40d25e14bd3f2efb574d81a12079b46adbbc06ca
Author: Jahnavi R <PES1UG24CS196>
Date:   1776770154

    Update file.txt

commit bdf433b863121575175810409a453904fcf5b170a88826c09874041868d2fd8b
Author: Jahnavi R <PES1UG24CS196>
Date:   1776770154

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
536d0316dfe49cc3931fd170f79e2a3f7cc06a26219a7c0fc4f4a02c98b75f55

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/38/42f48f44b8c0916a450c3f40d25e14bd3f2efb574d81a12079b46adbbc06ca
.pes/objects/53/6d0316dfe49cc3931fd170f79e2a3f7cc06a26219a7c0fc4f4a02c98b75f55
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
```

```
Author: Jahnavi R <PES1UG24CS196>
Date:   1776770154

    Add farewell

commit 3842f48f44b8c0916a450c3f40d25e14bd3f2efb574d81a12079b46adbbc06ca
Author: Jahnavi R <PES1UG24CS196>
Date:   1776770154

    Update file.txt

commit bdf433b863121575175810409a453904fcf5b170a88826c09874041868d2fd8b
Author: Jahnavi R <PES1UG24CS196>
Date:   1776770154

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
536d0316dfe49cc3931fd170f79e2a3f7cc06a26219a7c0fc4f4a02c98b75f55

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/38/42f48f44b8c0916a450c3f40d25e14bd3f2efb574d81a12079b46adbbc06ca
.pes/objects/53/6d0316dfe49cc3931fd170f79e2a3f7cc06a26219a7c0fc4f4a02c98b75f55
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/bd/f433b863121575175810409a453904fcf5b170a88826c09874041868d2fd8b
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ ^[[200~jahnavi1207@jahnavi1207-VirtualBox:~/PES1UG24CS196-pes-vcs$ make
```